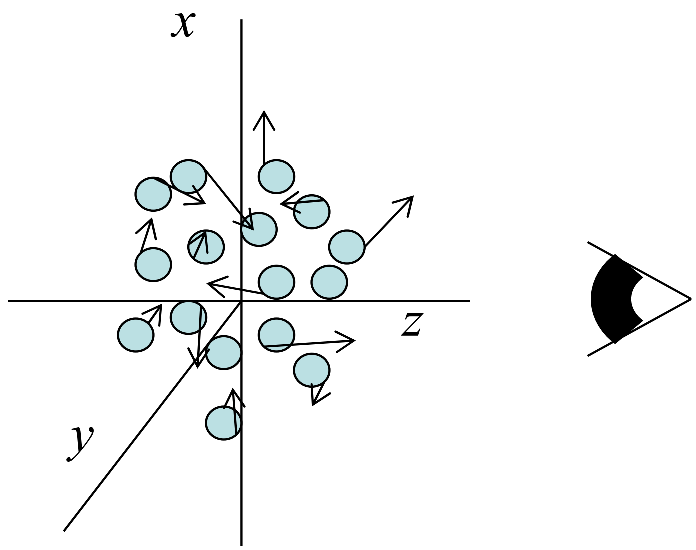
Ph134, Lecture 11

- **Some reasons for source bandwidth, and features of associated power spectra:**
  - **Doppler broadening**
  - **Natural linewidth of an atomic transition**
  - **Pressure broadening**

Source with continuous wavelength distribution? Doppler-broadened atomic transition



Atoms emit light at angular frequency ω_0 *in their own rest frames.*

From atom with v_z*, observer receives light at angular frequency*

$$\omega \approx \omega_0\left(1+\frac{\mathrm{v}_z}{c}\right)$$

Thermal spread in v_z *values* → *observed spread in* ω.

$s(\omega)$ from Doppler-broadened atomic transition

$$P(\mathrm{v}_z) = \frac{1}{\sqrt{2\pi\left(\frac{k_B T}{m}\right)}} e^{-\mathrm{v}_z{}^2/2\left(\frac{k_B T}{m}\right)}$$

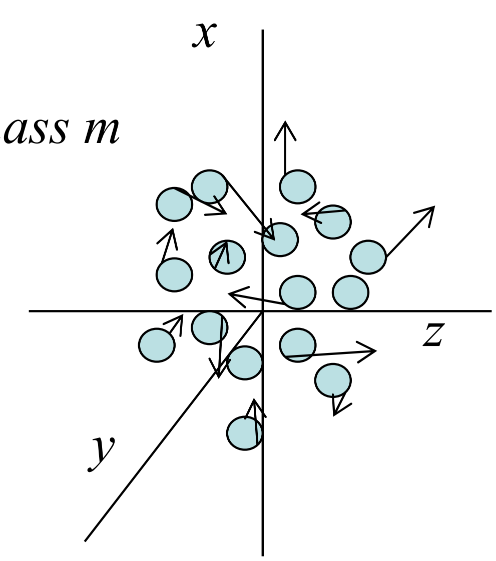
$$\omega \approx \omega_0\left(1 + \frac{\mathrm{v}_z}{c}\right)$$

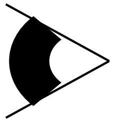
$$s(\omega)\, d\omega = P(\mathrm{v}_z)\, d\mathrm{v}_z$$

$$s(\omega) = \frac{1}{\sqrt{2\pi\sigma_\omega{}^2}} e^{-(\omega-\omega_0)^2/2\sigma_\omega{}^2}$$

$$\sigma_\omega{}^2 = \left(\frac{\omega_0}{c}\right)^2 \frac{k_B T}{m}$$

Atomic mass m

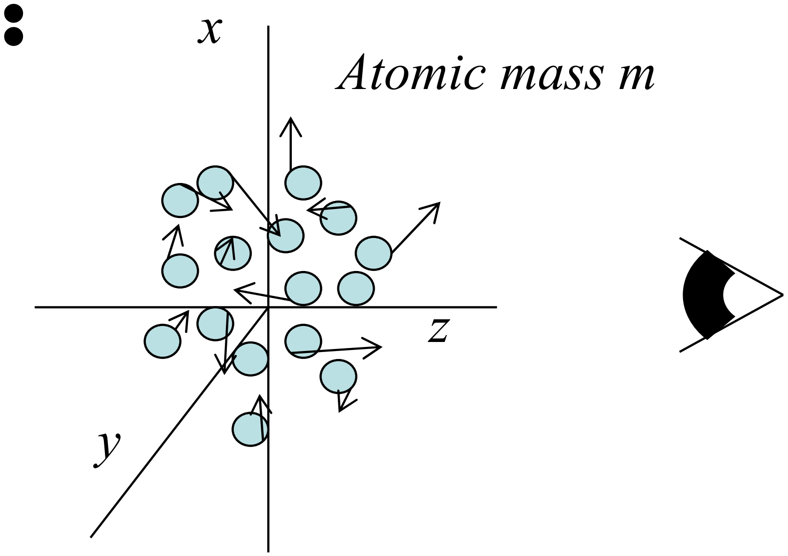




Doppler broadening examples:

$$s(\omega) = \frac{1}{\sqrt{2\pi\sigma_\omega{}^2}} e^{-(\omega-\omega_0)^2/2\sigma_\omega{}^2}$$

$$\sigma_\omega{}^2 = \left(\frac{\omega_0}{c}\right)^2 \frac{k_B T}{m}$$



Examples:

**He-Ne laser (633nm), Ne mass 20 amu, T~370 K: FWHM=2π*1.5GHz, or 0.02Å*

**solar H line (656nm), H mass 1 amu, T~5000K:FWHM=2π*24GHz, or 0.34Å due to Doppler alone*

...but pressure broadening actually dominates!

Interferometer output for a Doppler-broadened source?

From Fourier transform knowledge or by other means, find that $V(\tau)$ *is also Gaussian (centered at* $\tau = 0$*).*

What is FWHM and why are we talking about it?

FWHM: Full Width at Half Maximum

HWHM: Half Width at Half Maximum

For a Gaussian function $s(\omega) = \frac{1}{\sqrt{2\pi\sigma_\omega{}^2}} e^{-(\omega-\omega_0)^2/2\sigma_\omega{}^2}$:

$$s_{max} = s(\omega_0) = \frac{1}{\sqrt{2\pi\sigma_\omega{}^2}}$$

$$s(\omega) = \frac{1}{\sqrt{2\pi\sigma_\omega{}^2}} e^{-(\omega-\omega_0)^2/2\sigma_\omega{}^2} = \frac{1}{2} s_{max}$$

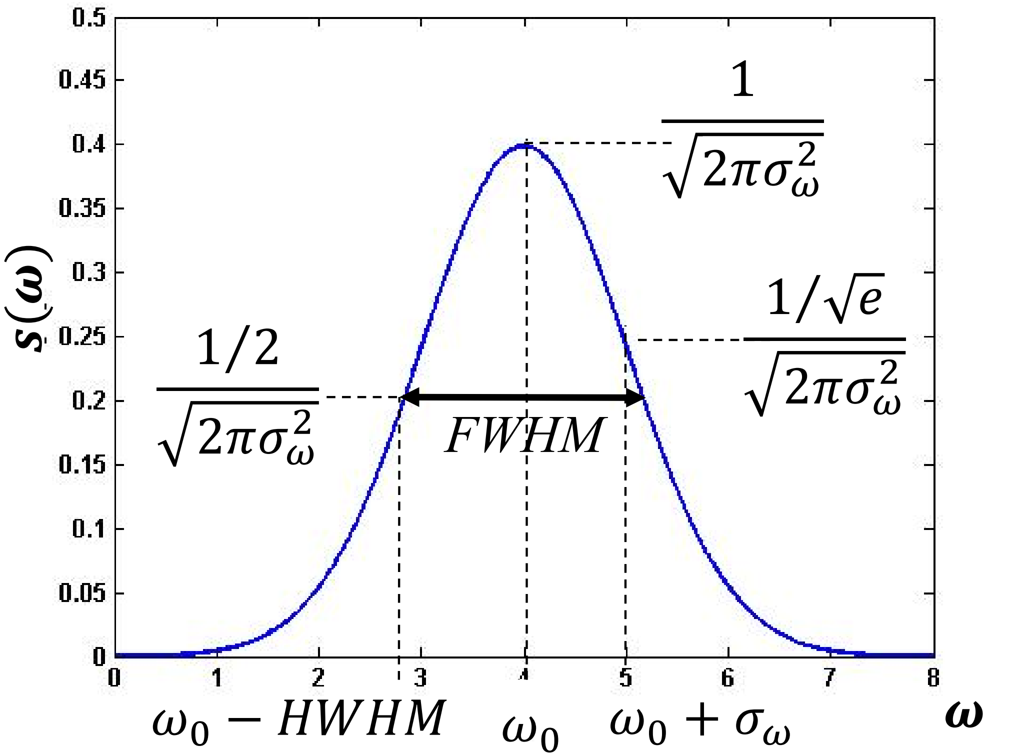
$$when \;\; e^{-(\omega-\omega_0)^2/2\sigma_\omega{}^2} = \frac{1}{2}$$

$$(\omega - \omega_0)^2/2\sigma_\omega{}^2 = \ln 2$$

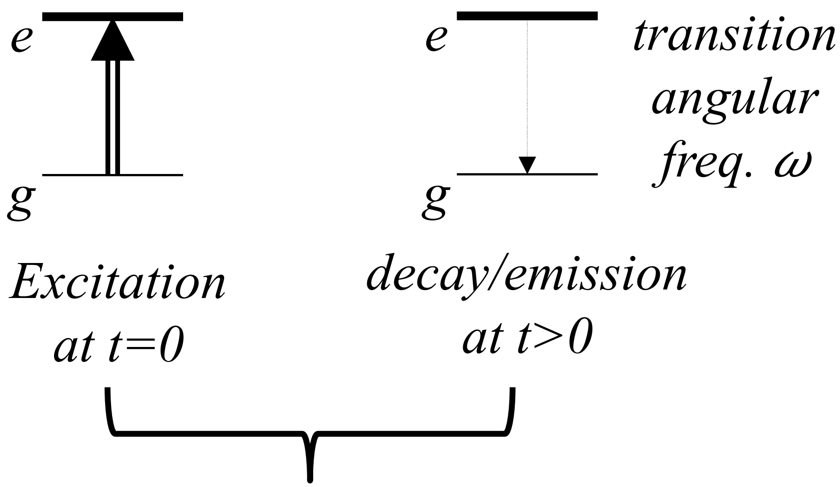
$$\omega = \omega_0 \pm \sqrt{2\ln 2}\,\sigma_\omega$$

$$HWHM = \sqrt{2\ln 2}\,\sigma_\omega \approx 1.177\sigma_\omega$$

$$FWHM = 2\sqrt{2\ln 2}\,\sigma_\omega \approx 2.35\sigma_\omega$$



Even an unbroadened atomic transition has some <u>natural linewidth</u>



Excited state mean lifetime T_L

$$\Delta Energy_e\, T_L \sim \hbar$$

$$\hbar(\Delta\omega)T_L \sim \hbar$$

$$\Delta\omega \sim \frac{1}{T_L}$$

Finite excited-state lifetime → *frequency spread of emitted light*

But exact form of function $s(\omega)$ *??*

Begin by finding form of $E(t)$ following a single excitation event:

Excitation at t=0

Spontaneous decay/emission at t>0

Initial excited population N_0

$$\frac{dN}{dt} = -C\,N(t)$$

$$N(t) = N_0 e^{-Ct}$$

$$\frac{dN}{dt} = -C N_0 e^{-Ct}$$

$$\langle t_{decay} \rangle = \frac{\int_0^\infty t\,(C N_0 e^{-Ct})\,dt}{\int_0^\infty (C N_0 e^{-Ct})\,dt} = \frac{1}{C}$$

$$\Rightarrow C = \frac{1}{T_L}$$

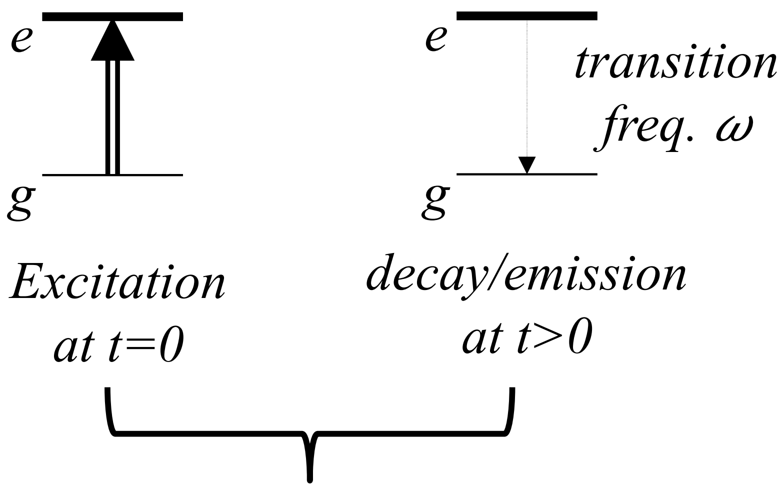
Excited state mean lifetime T_L

Initial excited population N_0

$$N(t) = N_0 e^{-t/T_L}$$

$$E^{(+)}(t) = E_0^{(+)} e^{-i\omega_0 t} e^{-t/2T_L}$$

$g^{(1)}(\tau)$ of natural linewidth source:

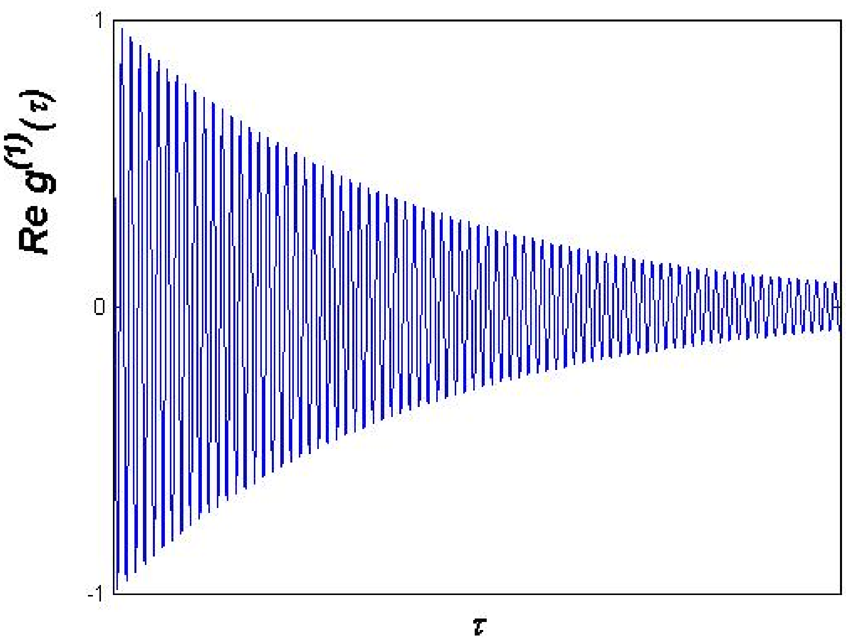


Excited state mean lifetime T_L

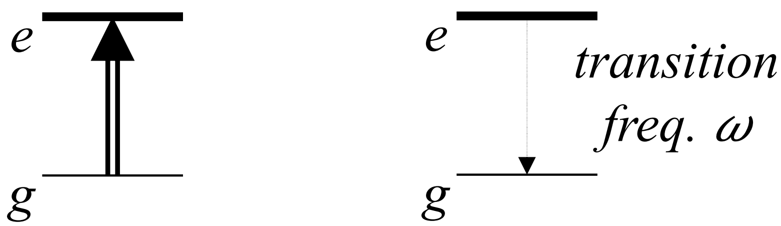
$$E^{(+)}(t) = E_0^{(+)} e^{-i\omega_0 t} e^{-t/2T_L}$$

$\omega_0 \sim 10^{15}$ rad/s $\qquad T_L \sim 10^{-8}$s

$$g^{(1)}(\tau) = e^{-i\omega_0 \tau} e^{-\tau/2T_L}$$



$s(\omega)$ **from Fourier transform of** $\mathrm{Re}\, g^{(1)}(\tau)$ **:**



Excitation at t=0

decay/emission at t>0

Excited state mean lifetime T_L

$$g^{(1)}(\tau) = e^{-i\omega_0\tau} e^{-\tau/2T_L}$$

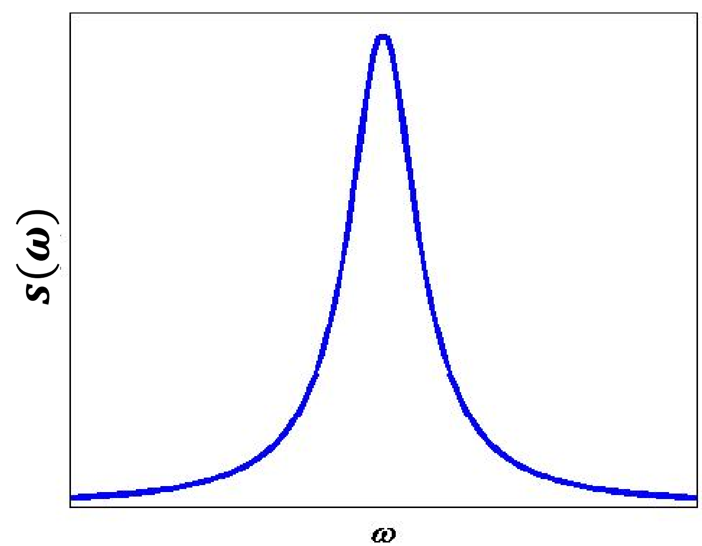
$\omega_0 \sim 10^{15}\ \mathrm{rad/s}$ $\qquad T_L \sim 10^{-8}\mathrm{s}$

$$s(\omega) = \frac{1}{\pi} \frac{\left(\frac{1}{2T_L}\right)}{(\omega-\omega_0)^2 + \left(\frac{1}{2T_L}\right)^2}$$

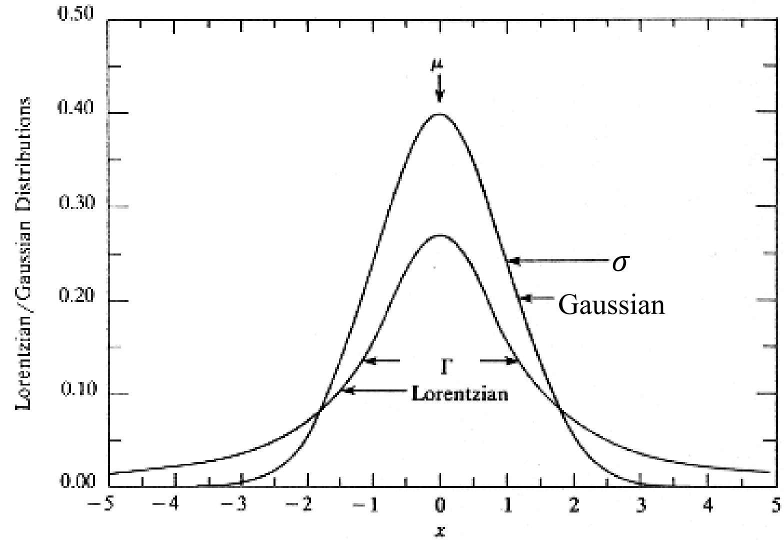
Lorentzian/Cauchy line shape

$\sigma_\omega = \infty$

FWHM$\approx {}^{1}/_{T_L}$ $\quad$ *(*$2\pi * 10^7$ Hz *typical)*



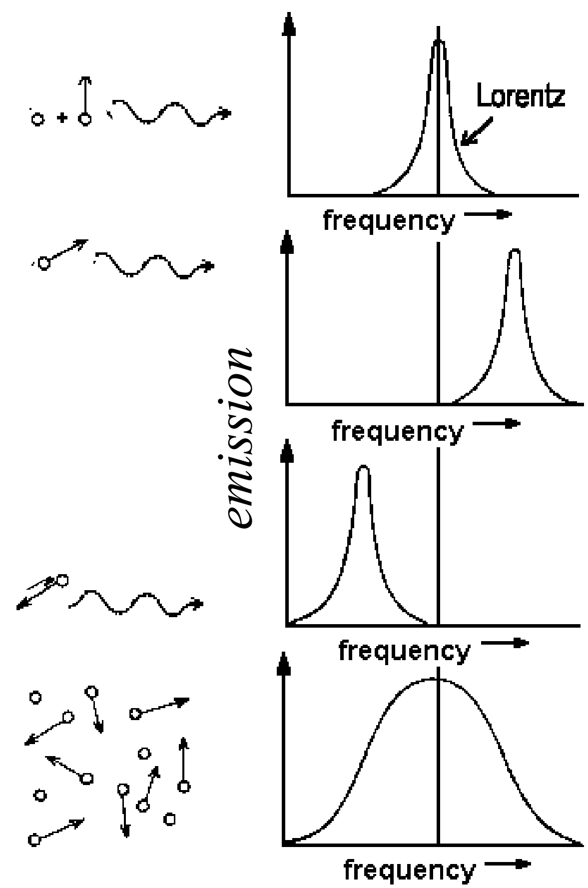
Gaussian vs. Lorentzian line shapes:



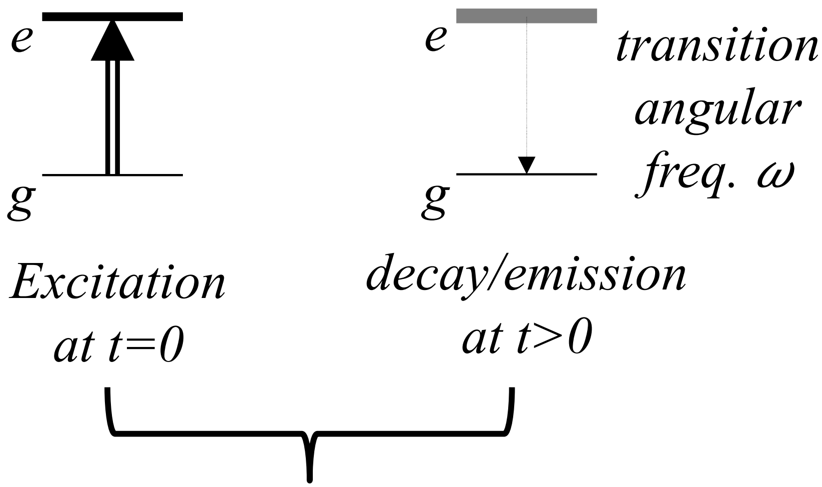
Doppler vs. natural linewidths:

**natural linewidth is <u>homogeneous</u>:
it is a characteristic of each and every individual atom in a sample*

**Doppler broadening is <u>inhomogeneous</u>:
its presence comes from different properties of different atoms in a sample*



Begin with a natural-linewidth transition:

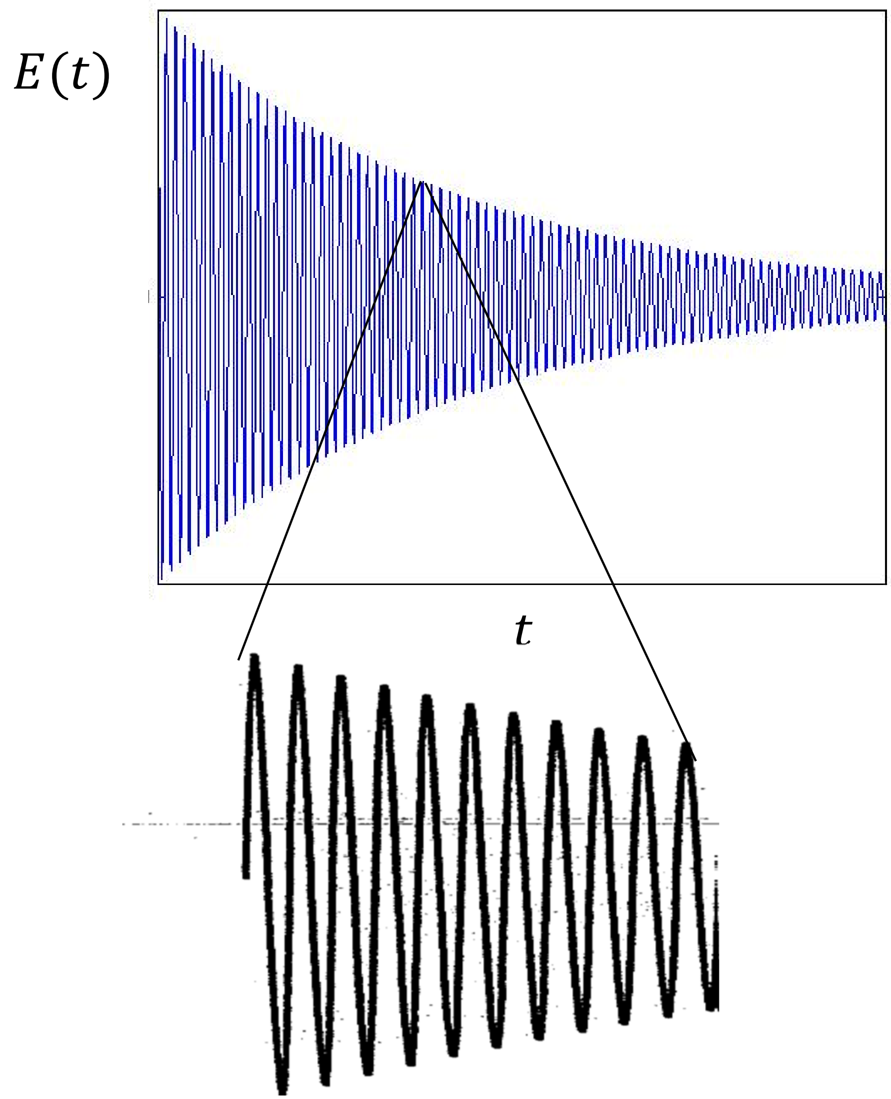


Excited state mean lifetime T_L

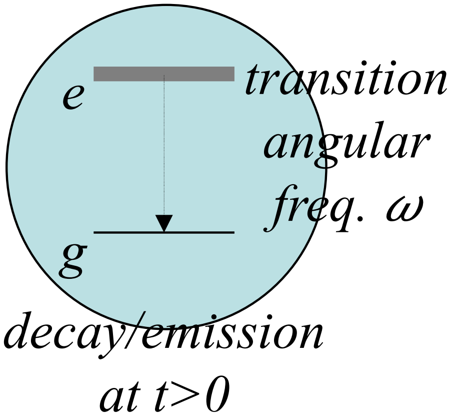
$$T_L \sim 10^{-8}\,\mathrm{s}$$

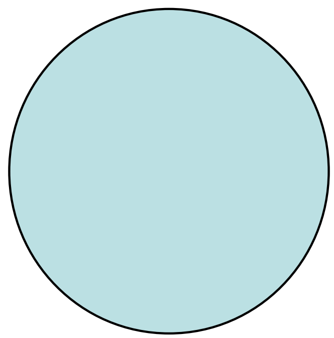
$$\omega_0 \sim 10^{15}\ \mathrm{rad/s}$$

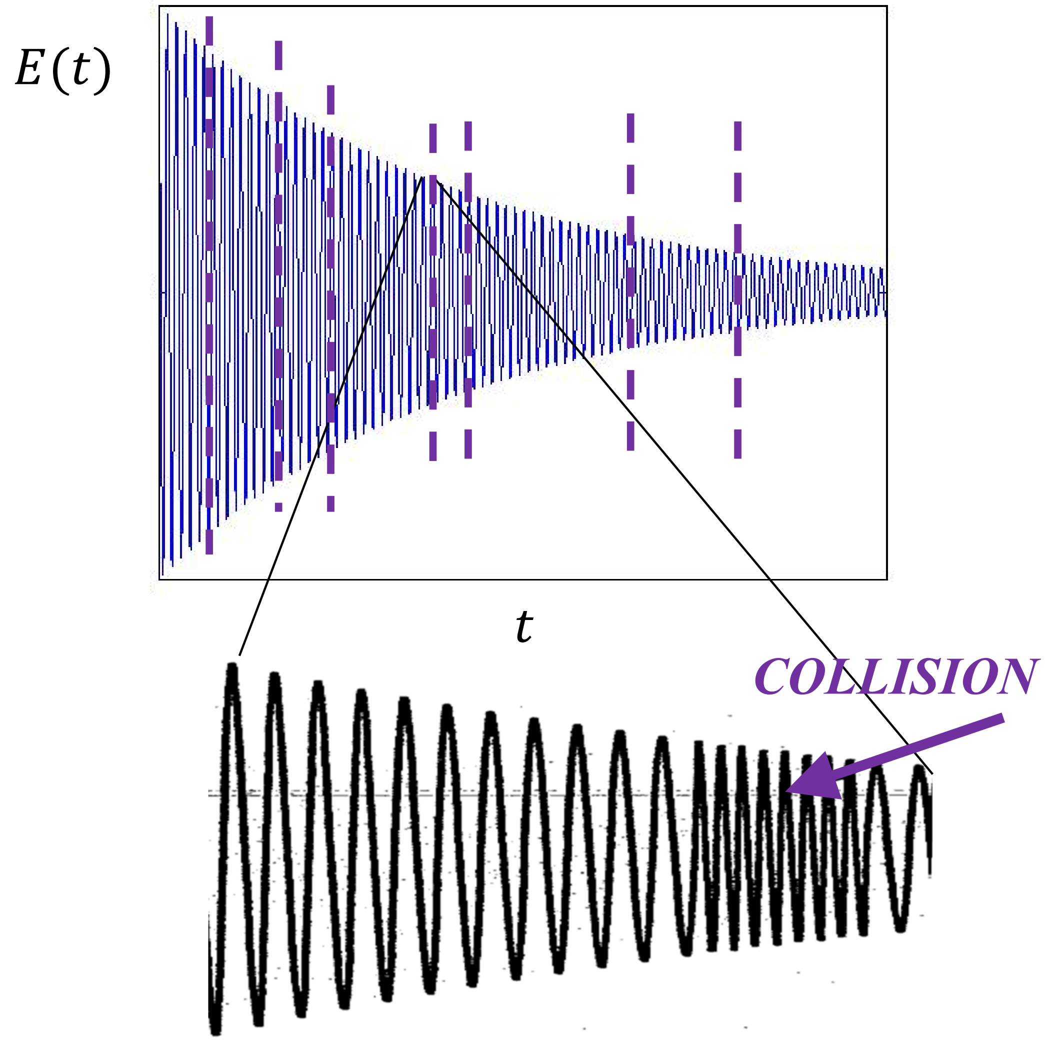
$$E^{(+)}(t) = E_0^{(+)} e^{-i\omega_0 t} e^{-t/2T_L}$$



Let atoms undergo collisions:







Excited state mean lifetime T_L

$$T_L \sim 10^{-8}\text{s}$$

$$\omega_0 \sim 10^{15}\ \text{rad/s}$$

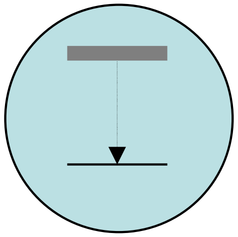
Collision duration δ

$$\delta \sim 10^{-13} - 10^{-12}\text{s}$$

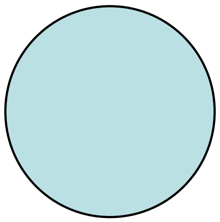
$$E^{(+)}(t) = E_0^{(+)} e^{-i(\omega_0 t + \varphi(t))} e^{-t/2T_L}$$

Randomized on time scale T_P*,*
mean time btwn collisions

Find $g^{(1)}(\tau)$ of pressure-broadened transition:

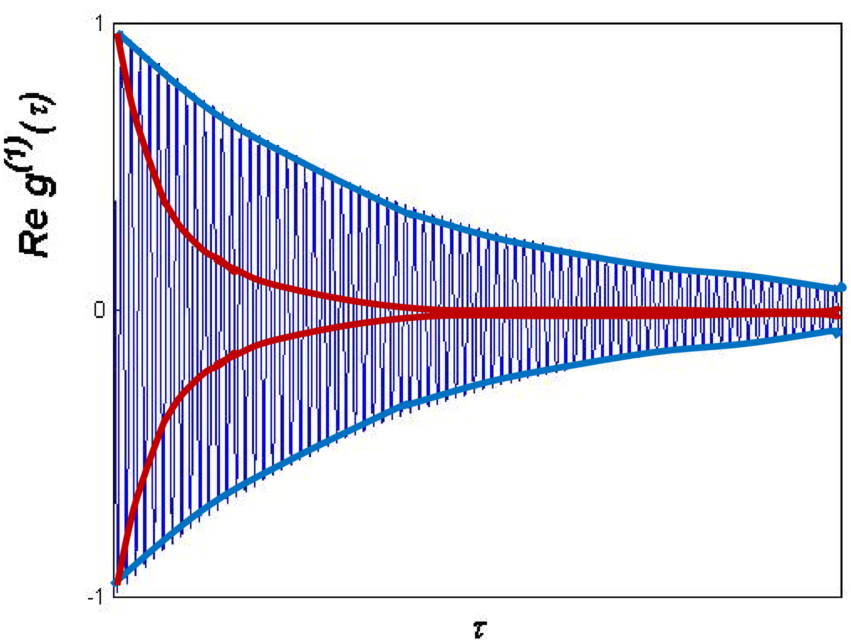


$$E^{(+)}(t) = E_0^{(+)} e^{-i(\omega_0 t + \varphi(t))} e^{-t/2T_L}$$

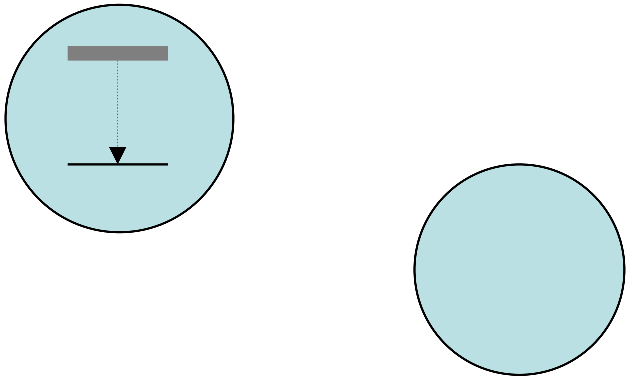


$$g^{(1)}(\tau) = e^{-\tau/2T_L} e^{-i\omega_0 \tau} \left\langle e^{-i[\varphi(t+\tau) - \varphi(t)]} \right\rangle$$

$$= e^{-\tau/2T_L} e^{-i\omega_0 \tau} e^{-\tau/T_P}$$



Find s(ω) of a pressure-broadened transition:

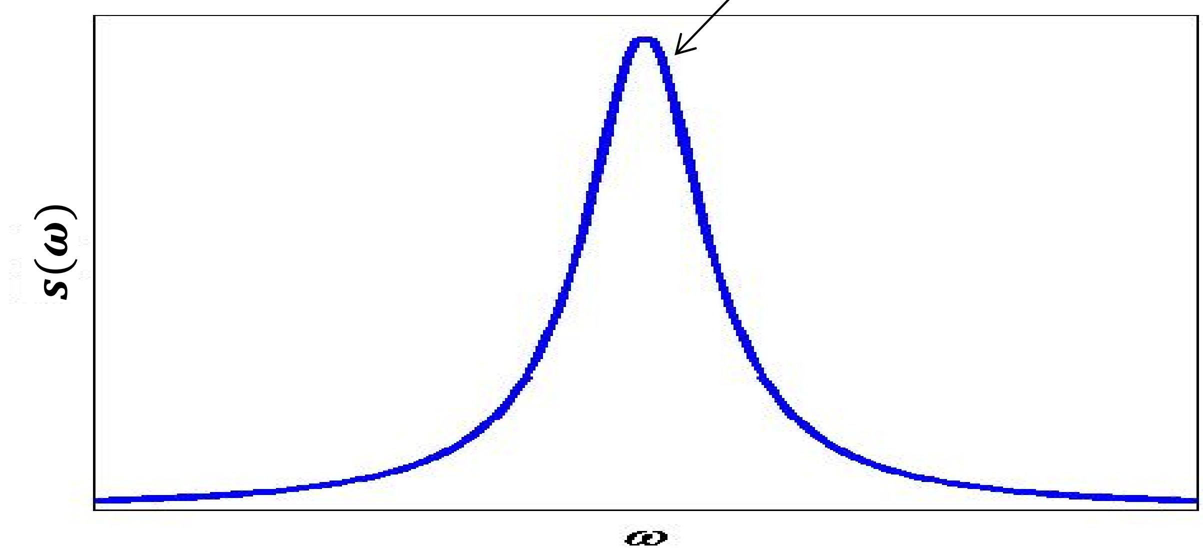


$$g^{(1)}(\tau) = e^{-\tau/2T_L} e^{-\tau/T_P} e^{-i\omega_0\tau}$$

$$s(\omega) = \frac{1}{\pi} \frac{\left(\frac{1}{2T_L}+\frac{1}{T_P}\right)}{(\omega-\omega_0)^2+\left(\frac{1}{2T_L}+\frac{1}{T_P}\right)^2}$$

May also have slight asymmetry about ω_0

Lorentzian lineshape



$FWHM \approx {}^{2}/_{T_P}$

$\sigma_\omega = \infty$